Nishant Gadde

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Education

The University of Texas at Austin — B.S. Electrical and Computer Engineering (Engineering Honors) and Mathematics | Expected May 2028 | GPA: 3.87/4.0

Experience

ML Researcher, Apple | Seattle, WA | Jun-Aug 2026

- Improved infrastructure for tracking ML experiments, helping researchers compare runs and study model behavior.
- Researched batching and caching strategies for large-scale inference workloads and analyzed throughput and latency tradeoffs.

Software Engineer, AI Security Infrastructure, Corveris | Spring 2026

- Worked on software engineering for AI security infrastructure.

Research Assistant, Vision-Language Models, Oden Institute, UT Austin | Austin, TX | Jan 2026-Present

- Developed affordance-centered vision-language methods for reasoning about object function and new concepts.
- Built evaluation and uncertainty-calibration workflows for model accuracy, latency, and concept acquisition.

ML and Quantitative Research Intern, Rayfield Systems | Houston, TX | Jun-Dec 2025

- Built recurrent state-space models for high-frequency time series and optimized streaming inference, reducing memory use 4x.
- Developed data pipelines, backtests, and risk monitoring for systematic power and gas trading strategies.
- Delivered model evaluation and drift monitoring for client-facing time-series applications.

Research Assistant, Stanford Sustainability Systems Laboratory | Remote | Jun 2024-Aug 2025

- Co-authored research on ML approximations of AC optimal power flow with <1% optimality gaps and 1000x faster solves.
- Evaluated feasibility with convex relaxations and benchmark power systems.

Selected Projects

NASDAQ ITCH Feed Handler | C++20, UDP multicast, SIMD | 2026

- Built a zero-allocation feed handler and live limit order book reconstruction pipeline.

Multi-Agent Research Assistant | Python, tool use, evaluation | 2025-2026

- Built planner, retrieval, execution, and review agents with persistent memory, citation tracking, and automated evaluations.

Selected Publications

- Siva et al., “What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning,” CoRL, 2026.
- Prakash et al., “QuArch: A Benchmark for Evaluating LLM Reasoning in Computer Architecture,” ICML, 2026.
- Miltner et al., “Towards Real-Time Machine Learning Approximations of AC Optimal Power Flow,” IEEE CAI, 2025.

Skills and Honors

Programming and systems: Python, C++, Rust, SQL, Linux, CUDA, Kafka, Git. **ML and quant:** PyTorch, TensorFlow, reinforcement learning, time-series modeling, Bayesian optimization, backtesting, risk analysis. **Honors:** USACO Platinum (top 1%); 3x AIME qualifier.